

OTC Derivatives Trade Reporting Compliance Engine

Recorded Presentation Deck

Project focus

A reproducible RegTech pipeline for OTC trade parsing, UPI taxonomy matching, CFTC / EMIR compliance checks, and EventContract classification-frontier analysis.

Presentation flow

1. Live demo → 2. One finding → 3. Policy argument → 4. Engine limitation

Team Members

Names and student IDs for the recorded presentation submission

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Deck structure: cover + four required segments in the exact order requested by the instructor.

Segment 1 — Live Demo

T017: EMIR Null Margin Failure

CFTC + EMIR | One-trade run

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1 **One-trade input**
output_demo/demo_T017.json



2 **Rule engine**
src/engine.py



3 **Outputs**
JSON + CSV + dashboard

> What we run

Command

```
python run_compliance_check.py
--input
output_demo/demo_T017.json
--regimes CFTC,EMIR
--output-dir output_demo/results
```

Demo objective:
prove the engine works on one concrete
trade and writes reproducible outputs.

✓ Observed result

trade_id	T017
overall_status	NONCOMPLIANT
data_quality_status	NONCOMPLIANT
reporting_scope_status	IN_SCOPE
regulatory_conclusion	CONVENTIONAL

Writes: compliance_report.json + findings.csv

! Key findings

EMIR_NULL_MARGIN

3 null margin fields

EMIR_REQUIRED_FIELD

missing collateral_portfolio_code

UPI status

WARNING

Why it matters

The trade is IN_SCOPE, but null margin fields are
reporting failures — zero must be reported as 0, not left
blank.

Narration focus

This is a conventional FX Forward: not outside the regime, not a parser failure — it is a reportable trade with clear EMIR data-quality defects.

Segment 2 — One finding

T027 exposes a UPI taxonomy gap

Full portfolio run | T027

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1 Full portfolio run
28 trades under CFTC + EMIR



2 Selected finding
T027: NO_PRODUCT_DEFINITION



3 Interpretation
taxonomy gap, not parser crash

i What T027 is

Trade **T027** EventContract

Asset class EventContract

Instrument BinaryOption

Use case EconomicIndicator

Platform Polymarket

Currency **USDC**

Prediction-market style event contract on an economic indicator.

! Observed result

UPI result **NO_PRODUCT_DEFINITION**

Overall **NOT_REPORTABLE**

Data quality **NONCOMPLIANT**

Parsed successfully, but no OTC UPI product template exists for EventContract.

? Why it matters

NO_PRODUCT_DEFINITION

No ANNA-DSB OTC UPI schema for EventContract.

INVALID_CURRENCY

USDC is not accepted as an ISO/DSB currency code.

LEI / UTI missing

The transaction cannot be linked cleanly to parties or identifiers.

Not bad parsing — a missing product taxonomy.

Narration focus

Taxonomy gap, not parser crash.

Segment 3 — Policy argument

T026–T028: classification frontier, not ordinary OTC swap failures

1

Policy claim

Create a dedicated category



2

Evidence

T026 / T027 / T028 split



3

Reason

avoid misclassification or invisibility

Policy claim

EventContracts should not be forced into ordinary OTC swap categories, but derivative-like EventContracts should receive a dedicated reporting category and UPI schema.

3

Classification matrix

Trade	Platform	Event type	Engine conclusion	Policy treatment
T026	Kalshi / CFTC DCM	PoliticalOutcome	CONDITIONAL_EVENT_CONTRACT	Conditional reporting candidate
T027	Polymarket / offshore	EconomicIndicator	NOT_REPORTABLE_EVENT_CONTRACT	Supervisory blind spot
T028	Kalshi / CFTC DCM	RegulatoryOutcome	CONDITIONAL_EVENT_CONTRACT	Conditional reporting candidate



Strongest reason

Current taxonomy creates a binary choice:

1

Misclassify EventContracts as ordinary OTC swaps

2

Exclude them entirely from reporting visibility

Dedicated category = visibility without false classification

Narration focus

Do not call every EventContract an ordinary swap; create a reporting path when the economic function is derivative-like.

Segment 4 — What the engine cannot see

Blind spot: aggregate EventContract exposure outside the current taxonomy

Limitation | Concrete risk

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1 Concrete actor
Polymarket-like offshore venue



2 Mechanism
no UPI + missing LEI/UTI



3 Risk
no aggregate event exposure view

! Concrete blind spot

3 / 28 = 10.7%

EventContracts in the sample

Actor: Polymarket-like offshore venue

Currency: USDC settlement

Identifiers: missing LEI / UTI

The engine can classify the gap; it cannot measure hidden market-wide concentration.

→ Mechanism

T027-like EventContract



No ANNA-DSB UPI product definition



Missing LEI / UTI + USDC settlement



Out of selected OTC reporting path



Regulator cannot aggregate event exposure

? What remains invisible

A taxonomy-based engine cannot estimate whether many T027-like contracts are building concentrated exposure to the same CPI event across venues.

Concrete risk:

- ✓ no standard product identifier
- ✓ no counterparty linkage
- ✓ no cross-platform aggregation

taxonomy-frontier problem, not just data-quality problem

Closing line

The engine can tell us T027 is outside the selected OTC path, but it cannot measure hidden aggregate exposure to the same macro event.